When Will a Borrower Default?

Survival Analysis in Credit Risk

Luke Munnelly

Why does timing matter?

- Loan A defaults after eight months.
- Loan B is still active after eighteen months.
- How can both observations inform an analysis?

Hypothetical loans.

Define the event and the clock

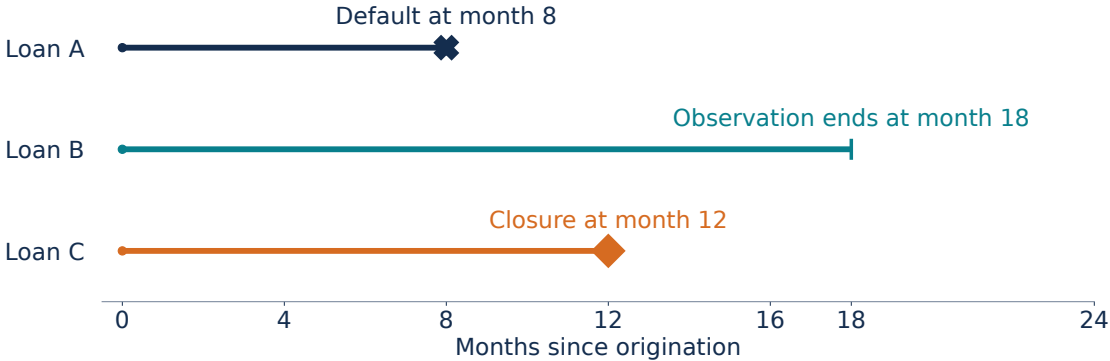
Origination $\longrightarrow$ first recorded default

Elapsed time since the loan was issued

An observed default or the end of observed exposure.

Loan timelines: three different endpoints

ILLUSTRATIVE · Hypothetical loans measured from origination



B: right censoring — still active when observation ends.

C: closure — a competing event that ends default exposure.

The survival function

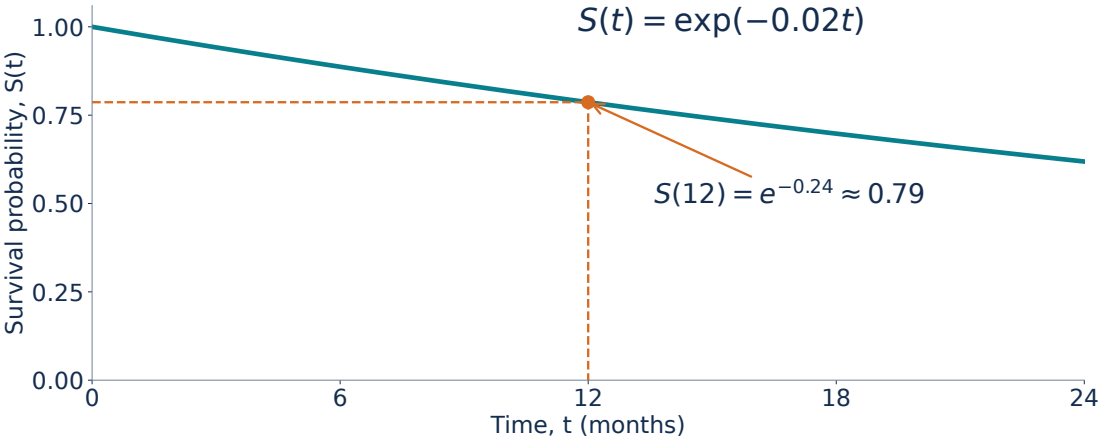
First consider a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The probability of remaining default-free beyond time t.

Survival: the probability of remaining default-free

HYPOTHETICAL · Default-only setting; not a Bondora estimate



At 12 months, about 79% remain default-free in this hypothetical setting.

The hazard is an instantaneous rate

Hypothetical default-only setting; t is in months.

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \le T_D < t + \Delta t \mid T_D \ge t)}{\Delta t}$$

$$P(t \le T_D < t + \Delta t \mid T_D \ge t) \;\approx\; h(t)\,\Delta t$$

Units: per month. A rate, not a probability.

From hazard to survival

Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

$$S(t) = \exp\{-H(t)\}$$

Competing closure: default hazard alone does not determine actual default probabilities.

The Cox model

$$h_D(t \mid x, y) = \underbrace{h_{0D,y}(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{relative effect}}$$

A cause-specific default rate among loans still exposed.

y: origination year. Baseline hazards may differ by year.

A hazard ratio compares two rates

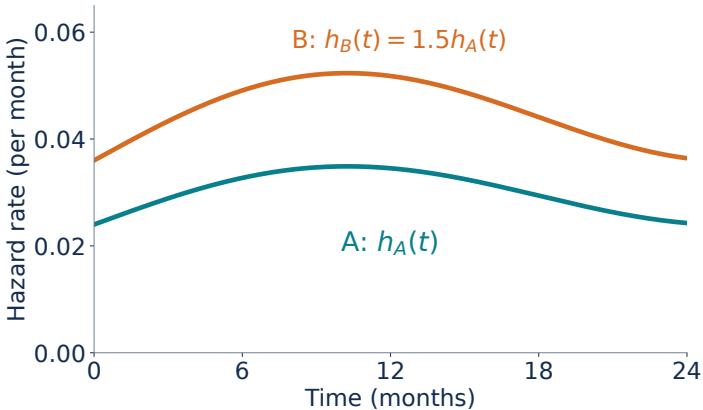
$$\frac{h_D(t \mid x + \Delta x, y)}{h_D(t \mid x, y)} = \exp(\beta^{\mathsf{T}} \Delta x)$$

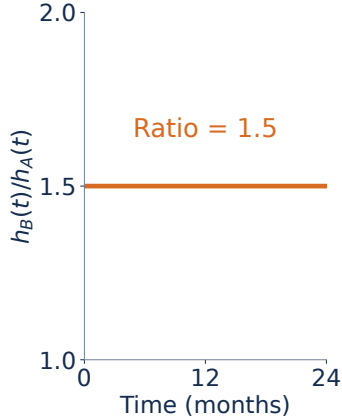
Under PH, the ratio does not depend on t.

Compare loans in the same year stratum. A hazard ratio is not a default-probability ratio.

Proportional hazards: changing rates, constant ratio

HYPOTHETICAL · A hazard is an instantaneous rate, not a probability





The baseline hazard can vary over time; proportionality concerns the ratio.

Immediately before one default

Illustrative risk set; all other covariates equal.

A	B	C
Existing customer	Existing customer	New customer
Relative risk 1	Relative risk 1	Relative risk e^{β}

Which borrower accounts for this default?

Cox regression compares borrowers in the risk set

ILLUSTRATIVE · At risk immediately before a default; all other covariates equal

A	B	C
Existing customer	Existing customer	New customer
Relative risk: 1	Relative risk: 1	Relative risk: e^{β}

$$P(C \text{ defaults} \mid \text{one default in risk set}) = \frac{e^{\beta}}{2 + e^{\beta}}$$

If $\beta = \log(2)$: $\quad \frac{2}{1+1+2} = \frac{1}{2}$

Conditional on one default occurring now; not a probability over a time horizon.

The audited Bondora cohort

Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

9,844	6,924	2,508
Loans	Borrowers	Recorded defaults

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

Follow-up uses audited endpoints

- Default: first recorded `DefaultDate`.
- Closure proxy: `ContractEndDate` only when snapshot `Status=Repaid`.
- Stop at the first exit or 730 days.

Closure is not guaranteed full cash repayment. The source default threshold is unverified.

Five covariates; one Cox model

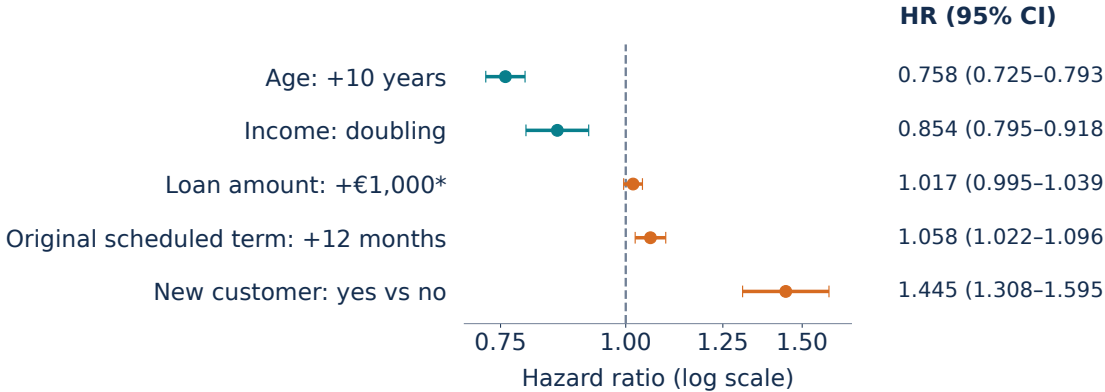
- Age; log income.
- Amount; original scheduled term.
- New-customer status.

Year-specific baseline hazards; borrower-clustered intervals.

Original term is reconstructed from the original monthly schedule. Monetary values are used as recorded.

Recorded default: adjusted hazard ratios

EMPIRICAL · Final audited Cox estimates; baseline stratified by origination year



Borrower-clustered 95% CIs are approximate with tied daily event times.

PH departures: age and new-customer status. Nonlinearity: age; borderline for term.

*Monetary fields as recorded; source currency/period definitions remain unverified.

Conditional associations, not causal effects or default probabilities. See REPORT.md.

New versus existing customer

$$\text{HR} = 1.445$$

95% CI: 1.308–1.595

About 44% higher conditional default rate among loans still exposed.

Association, not a default-probability ratio or causal effect.

PH concern: this single HR is a model summary.

Where interpretation is limited

- PH concerns: age and new-customer status.
- Nonlinearity and approximate tied-date intervals.
- Recorded endpoints and measurement proxies.

Conclusion

- Survival analysis uses timing and incomplete follow-up.
- Cox describes default-rate associations among exposed loans.

Endpoint definitions and model assumptions shape the interpretation.

Backup: partial likelihood

$$L(\beta) = \prod_{j:\,\text{default}} \frac{\exp(\beta^{\mathsf{T}} x_{i_j})}{\sum_{k \in \mathcal{R}_{y_j}(t_j)} \exp(\beta^{\mathsf{T}} x_k)}$$

$\mathcal{R}_{y_j}(t_j)$: loans still exposed immediately before t_j, within the defaulting loan's origination-year stratum.

Schematic formula for distinct event times. The audited fit uses Efron handling of ties.

Backup: corrected diagnostics

	PH rank p	PH log p	Nonlinear p
Age	0.00641	0.00768	0.00358
Log income	0.933	0.621	0.185
Amount	0.236	0.118	0.805
Original term	0.777	0.997	0.0438
New customer	0.00888	0.000417	—

PH: approximate independence-based Schoenfeld tests, not cluster-adjusted. Rank uses elapsed event times. Nonlinearity: existing spline cluster-Wald checks; exploratory.

One loan per borrower: 6,924 loans, 1,842 defaults. Broadly similar coefficient directions; sample composition changes.